

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
BOIN12	36.1	14.4	5.9	1.5	0.7	41.4
U-BOIN	27.8	20.4	16	13.4	9.5	12.9
EffTox	5	4	6	3	6	75
AIDE one_stage (U2)	33.6	15.6	1.9	0.6	0.2	48.1
AIDE one_stage (U3)	35	14.1	1.5	0.2	0	49.2
AIDE two_stage (U2)	29.6	10.2	1.5	0.4	0.2	58.1
AIDE two_stage (U3)	31.7	9.2	0.5	0.2	0.2	58.2
Mean patients treated						
BOIN12	13.5	6	3.2	1.9	0.9	
U-BOIN	9.8	5.4	4.3	3.8	3.6	
EffTox	8	3.2	3	2	2	
AIDE one_stage (U2)	14.6	6.2	2.4	1.4	0.4	
AIDE one_stage (U3)	15.1	5.7	2.3	1.3	0.4	
AIDE two_stage (U2)	15.4	3.9	2.2	0.9	0.3	
AIDE two_stage (U3)	15.4	3.8	2.1	0.9	0.3	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
BOIN12	29.2	24.7	33.9	9.2	1.5	1.5
U-BOIN	11.2	11.5	40.6	24.1	11.7	0.9
EffTox	72	7	5	6	9	1
AIDE one_stage (U2)	26.3	19.2	38.3	12.8	2.2	1.2
AIDE one_stage (U3)	31.3	20.1	35.7	10.6	1.3	1
AIDE two_stage (U2)	36.2	16.3	32.4	9.7	3.6	1.8
AIDE two_stage (U3)	38.8	17	31.4	8.6	2.4	1.8
Mean patients treated						
BOIN12	8.6	7.8	8.6	3.6	1.1	
U-BOIN	8.7	6.9	8.2	4.3	1.8	
EffTox	22.6	2.5	1.6	1.2	1.7	
AIDE one_stage (U2)	9.4	7.1	8.9	3.6	0.8	
AIDE one_stage (U3)	10	7.2	8.7	3.4	0.7	
AIDE two_stage (U2)	11.5	5.2	7.2	3.9	1.9	
AIDE two_stage (U3)	11.5	5.3	7.2	3.9	1.9	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
BOIN12	13.2	56	20.6	6.9	1.2	2.1
U-BOIN	3.8	44.7	24.5	16.2	10	0.8
EffTox	49	19	8	7	15	2
AIDE one_stage (U2)	9.7	50.1	23.7	11.3	2.5	2.7
AIDE one_stage (U3)	11.7	53.6	21.8	8.5	1.5	2.9
AIDE two_stage (U2)	18.7	40	21	7.6	2.4	10.3
AIDE two_stage (U3)	19.6	42.5	19.2	6.6	1.9	10.2
Mean patients treated						
BOIN12	6.9	11.3	6.4	3.5	1.6	
U-BOIN	7.2	10.7	6.7	3.6	1.7	
EffTox	18.8	4.8	2.6	1.5	1.9	
AIDE one_stage (U2)	6.8	11.7	6.8	3.5	1	
AIDE one_stage (U3)	6.9	12.4	6.4	3.2	0.8	
AIDE two_stage (U2)	9.7	7.6	6.2	3.6	1.8	
AIDE two_stage (U3)	9.7	7.6	6.2	3.6	1.7	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
BOIN12	15.6	14.2	25.6	36.9	7.4	0.3
U-BOIN	3.7	6	18.4	43.5	28.4	0
EffTox	7	14	20	17	40	1
AIDE one_stage (U2)	6.6	6.1	17	54.5	15.5	0.3
AIDE one_stage (U3)	10.4	8.5	16.3	52.6	11.9	0.3
AIDE two_stage (U2)	7.4	7.6	21	44.4	17.7	1.9
AIDE two_stage (U3)	11.8	9.7	19.4	42.5	14.7	1.9
Mean patients treated						
BOIN12	5.4	5.9	6.9	8.7	3.1	
U-BOIN	5.1	6.1	7.4	7.7	3.9	
EffTox	9.4	4.8	5.3	3.7	6.4	
AIDE one_stage (U2)	4.3	4.7	6.8	10.7	3.6	
AIDE one_stage (U3)	4.5	4.7	6.8	10.7	3.3	
AIDE two_stage (U2)	4.5	4.3	6.7	8.1	6.1	
AIDE two_stage (U3)	4.6	4.4	6.6	8.2	6	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
BOIN12	7	42.3	25.2	18	7	0.5
U-BOIN	2.9	37.5	26.5	21.3	11.7	0.1
EffTox	22	22	17	13	25	2
AIDE one_stage (U2)	5.8	39.4	24.5	20.2	9.5	0.6
AIDE one_stage (U3)	9.1	43.6	24	16.7	6	0.6
AIDE two_stage (U2)	4.4	38	24.9	19.3	9.1	4.3
AIDE two_stage (U3)	7.6	42.3	23.7	15.5	6.6	4.3
Mean patients treated						
BOIN12	4.5	7.8	6.7	6.7	4.4	
U-BOIN	5.2	8.5	6.9	5.7	3.9	
EffTox	12.8	5.9	4.7	2.8	3.5	
AIDE one_stage (U2)	3.9	7.5	7.2	7	4.4	
AIDE one_stage (U3)	4	8.2	7.2	6.8	3.8	
AIDE two_stage (U2)	4.9	6.7	6.6	5.9	5.3	
AIDE two_stage (U3)	5	6.6	6.6	5.9	5.4	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
BOIN12	64.3	16.7	5.3	1.7	0.4	11.6
U-BOIN	25.6	17.2	13.9	17.6	14.3	11.4
EffTox	85	1	2	1	1	10
AIDE one_stage (U2)	62.6	17.7	4.7	2	0.4	12.6
AIDE one_stage (U3)	65.2	17.2	3.1	1.5	0.4	12.6
AIDE two_stage (U2)	68.4	12.5	3.8	1	0.5	13.8
AIDE two_stage (U3)	69.1	12.9	3.3	0.7	0.4	13.6
Mean patients treated						
BOIN12	15.9	6.4	3.3	1.8	0.6	
U-BOIN	11.9	6.1	4.3	3.1	1.9	
EffTox	25.6	1.1	0.6	0.3	0.3	
AIDE one_stage (U2)	17.6	6.1	2.7	1.2	0.3	
AIDE one_stage (U3)	18.1	5.9	2.4	1.1	0.2	
AIDE two_stage (U2)	19.5	4	2.5	1.1	0.4	
AIDE two_stage (U3)	19.5	4	2.5	1.1	0.4	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
BOIN12	12.8	51.1	20.7	9.8	3.4	2.2
U-BOIN	4.3	48.3	25.8	13.7	6.2	1.7
EffTox	72	8	6	4	10	1
AIDE one_stage (U2)	12.1	51.1	22.7	9.9	3.1	1.1
AIDE one_stage (U3)	14.2	54.2	20.2	8.7	1.5	1.2
AIDE two_stage (U2)	27	36.9	20.5	9.8	3.5	2.3
AIDE two_stage (U3)	28.9	40.5	18.4	8	2	2.2
Mean patients treated						
BOIN12	6.9	12.5	5.9	3.1	1.2	
U-BOIN	7.6	10.7	6.5	3.4	1.4	
EffTox	22.9	2.8	1.6	1.1	1.4	
AIDE one_stage (U2)	7.7	13.1	5.7	2.7	0.8	
AIDE one_stage (U3)	8	13.7	5.2	2.4	0.6	
AIDE two_stage (U2)	10.8	7.3	5.6	3.7	2.3	
AIDE two_stage (U3)	10.8	7.2	5.6	3.7	2.3	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
BOIN12	46.2	15	5.1	1.6	0.5	31.6
U-BOIN	14.8	10.2	14.5	17.4	26.7	16.4
EffTox	27	11	8	5	9	39
AIDE one_stage (U2)	44.7	14.3	3	1	0.2	36.8
AIDE one_stage (U3)	45.8	14	2.1	0.8	0.2	37.1
AIDE two_stage (U2)	42.5	9.4	1.9	1.2	0.3	44.7
AIDE two_stage (U3)	44.8	7.9	1.6	0.8	0.3	44.6
Mean patients treated						
BOIN12	14.5	5.9	3.1	1.7	0.7	
U-BOIN	8.9	4.7	4.2	4.3	4.1	
EffTox	14.3	3.3	2.4	1.5	1.9	
AIDE one_stage (U2)	15.7	6.1	2.3	1.1	0.3	
AIDE one_stage (U3)	16.1	5.9	2.2	1.1	0.2	
AIDE two_stage (U2)	17.4	3.7	1.9	0.8	0.2	
AIDE two_stage (U3)	17.4	3.7	1.9	0.7	0.2	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
BOIN12	18.9	16.2	22.4	12.6	25.3	4.6
U-BOIN	3.8	3.9	13.8	19.1	58	1.4
EffTox	8	8	12	8	56	9
AIDE one_stage (U2)	16.4	15.3	20.8	16.4	25.8	5.3
AIDE one_stage (U3)	22.1	17.8	17.9	14	22.1	6.1
AIDE two_stage (U2)	14.7	10.8	18	12.9	24.1	19.5
AIDE two_stage (U3)	20.1	11.9	16.7	11.7	20.3	19.3
Mean patients treated						
BOIN12	6.6	5.9	5.7	5.2	6.2	
U-BOIN	6.8	5.6	6.3	5.2	6	
EffTox	9.3	3.9	4.6	3	7.7	
AIDE one_stage (U2)	7.1	6.2	6	5.5	4.8	
AIDE one_stage (U3)	7.7	5.9	5.8	5.3	4.6	
AIDE two_stage (U2)	8.9	4.4	5.6	4.1	4.8	
AIDE two_stage (U3)	8.9	4.4	5.5	4.1	4.8	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
BOIN12	14.6	43.9	16.1	4	2.2	19.2
U-BOIN	1.2	32.8	27.1	17	17.3	4.6
EffTox	22	19	10	12	29	8
AIDE one_stage (U2)	12.4	42	15.3	6.7	1.3	22.3
AIDE one_stage (U3)	13.1	44.3	14.2	4.9	0.8	22.7
AIDE two_stage (U2)	11.9	30.8	12	3.6	1.8	39.9
AIDE two_stage (U3)	12.7	31.3	12	2.8	1.5	39.7
Mean patients treated						
BOIN12	8.8	10.4	5.1	2.5	1.1	
U-BOIN	7.7	9.2	6.5	3.6	2	
EffTox	14.5	5.4	3.2	2.2	3.2	
AIDE one_stage (U2)	9.7	10.6	4.7	2.1	0.6	
AIDE one_stage (U3)	9.9	11	4.4	1.9	0.5	
AIDE two_stage (U2)	11.8	6	4	2.1	1	
AIDE two_stage (U3)	11.8	6	4.1	2.1	1	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
BOIN12	64.7	23.1	4.4	0.2	0.1	7.5
U-BOIN	49.1	19.1	13.5	6.8	8	3.5
EffTox	69	5	3	2	1	20
AIDE one_stage (U2)	66.3	23.4	4.1	0.4	0	5.8
AIDE one_stage (U3)	69.8	20.5	3.7	0.4	0	5.6
AIDE two_stage (U2)	66.8	20.9	4.3	0.9	0	7.1
AIDE two_stage (U3)	69.4	20	2.8	0.6	0	7.2
Mean patients treated						
BOIN12	14.7	8	3.9	1.7	0.5	
U-BOIN	11.9	6.6	4.7	3.2	2.8	
EffTox	21.8	2.2	1.5	0.8	0.4	
AIDE one_stage (U2)	16.7	7.8	3.3	1.3	0.2	
AIDE one_stage (U3)	17.3	7.5	3.1	1.2	0.2	
AIDE two_stage (U2)	18.5	5.9	3.5	1	0.1	
AIDE two_stage (U3)	18.4	5.9	3.6	1	0.1	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
BOIN12	20	26.2	13.1	31	9.5	0.2
U-BOIN	6.1	9.6	6.3	38.7	39.2	0.1
EffTox	10	9	24	18	36	4
AIDE one_stage (U2)	12.6	12.7	7.7	44.5	22.1	0.4
AIDE one_stage (U3)	16.9	14.9	7.7	40.1	20.1	0.3
AIDE two_stage (U2)	15	14.2	9.2	36.5	21.8	3.3
AIDE two_stage (U3)	22	13.7	6.9	34.6	19.6	3.2
Mean patients treated						
BOIN12	5.4	6.5	5.9	7.9	4.2	
U-BOIN	5.2	6.3	6.4	7.4	4.9	
EffTox	9.6	4.1	6.1	3.9	5.8	
AIDE one_stage (U2)	4.6	5.3	5.6	9.5	5.1	
AIDE one_stage (U3)	4.9	5.3	5.4	9.6	4.8	
AIDE two_stage (U2)	5.1	4.8	5.4	7.5	6.8	
AIDE two_stage (U3)	5.1	4.7	5.4	7.6	6.8	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
BOIN12	18.3	41.2	17.5	5	0.9	17.1
U-BOIN	4.3	21.5	27.2	21.4	21.3	4.3
EffTox	27	17	11	11	24	10
AIDE one_stage (U2)	14	43.3	15	6.3	1.7	19.7
AIDE one_stage (U3)	15.8	44	14.4	4.7	1.3	19.8
AIDE two_stage (U2)	14.1	31.1	13.8	4.3	1.8	34.9
AIDE two_stage (U3)	14.6	32.5	12.5	4	1.4	35
Mean patients treated						
BOIN12	9.5	9.1	5.3	3	1.3	
U-BOIN	7.9	7.1	6.5	4.5	3	
EffTox	14.5	5.4	3.2	2	2.8	
AIDE one_stage (U2)	9.7	10	4.8	2.7	0.8	
AIDE one_stage (U3)	9.9	10.1	4.7	2.6	0.7	
AIDE two_stage (U2)	11.4	6.4	4.5	2.4	1.1	
AIDE two_stage (U3)	11.4	6.3	4.5	2.3	1.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
BOIN12	28.7	22.2	13.5	24.9	9	1.7
U-BOIN	11.9	9.5	9.2	33.9	35.1	0.4
EffTox	9	9	21	15	36	11
AIDE one_stage (U2)	21.1	17.9	12.9	31.8	15.4	0.9
AIDE one_stage (U3)	27.2	20.7	11.1	28.7	11	1.3
AIDE two_stage (U2)	21	16	13	27.9	14.5	7.6
AIDE two_stage (U3)	31.7	17.4	8.5	24.3	10.6	7.5
Mean patients treated						
BOIN12	6.6	6.8	5.8	6.6	4.1	
U-BOIN	6.5	6.6	5.9	6.3	4.8	
EffTox	9.4	3.9	5.8	3.6	5.8	
AIDE one_stage (U2)	6.2	6.2	5.7	7.7	4.1	
AIDE one_stage (U3)	6.9	6.1	5.4	7.7	3.8	
AIDE two_stage (U2)	6.7	5.4	6	6.2	4.7	
AIDE two_stage (U3)	6.8	5.5	5.8	6.2	4.7	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
BOIN12	49.4	30.8	12.2	3.4	1.3	2.9
U-BOIN	41.1	23.2	11.8	11.2	12.2	0.5
EffTox	26	11	10	9	7	37
AIDE one_stage (U2)	48.8	31.3	11.6	5.5	0.6	2.2
AIDE one_stage (U3)	58.1	28.1	8.2	3	0.3	2.3
AIDE two_stage (U2)	54	23.5	10	4.2	0.9	7.4
AIDE two_stage (U3)	63.9	22	4.9	1.5	0.3	7.4
Mean patients treated						
BOIN12	10.3	8.4	5.5	3.6	2	
U-BOIN	9	7.6	5.6	4.3	3.5	
EffTox	14.1	3.2	3.7	2.1	1.8	
AIDE one_stage (U2)	10.2	8.4	5.7	4	1.5	
AIDE one_stage (U3)	11.2	8	5.3	3.8	1.4	
AIDE two_stage (U2)	11.9	6.7	5.9	3.2	1.4	
AIDE two_stage (U3)	11.7	6.8	5.9	3.2	1.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
BOIN12	36.8	26.2	20.4	11.7	4.9	0
U-BOIN	30.4	21.6	22.9	17.1	8	0
EffTox	66	5	8	7	9	5
AIDE one_stage (U2)	40.7	24.2	15.2	13.6	6.3	0
AIDE one_stage (U3)	46.6	23.3	15.1	10.4	4.6	0
AIDE two_stage (U2)	31.1	23.7	17.4	16.2	11.5	0.1
AIDE two_stage (U3)	38.9	24.9	15.7	13.1	7.3	0.1
Mean patients treated						
BOIN12	5	5.7	6.5	7.4	5.4	
U-BOIN	6.1	6	6.2	6.4	5.6	
EffTox	20.9	2.3	2.5	1.7	2.1	
AIDE one_stage (U2)	4.4	5	6	8.4	6.2	
AIDE one_stage (U3)	4.6	5.2	6.2	8.3	5.7	
AIDE two_stage (U2)	5.8	5.7	5.5	5.9	7	
AIDE two_stage (U3)	5.9	5.6	5.4	6	7	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
BOIN12	33.7	55.7	9.2	0.7	0	0.7
U-BOIN	28.7	50.6	13.6	4.4	2.4	0.3
EffTox	91	5	3	1	0	1
AIDE one_stage (U2)	19.4	61.8	16.8	1.7	0.1	0.2
AIDE one_stage (U3)	23.3	63	12.2	1.4	0	0.1
AIDE two_stage (U2)	30.4	50.7	16.8	1.7	0.1	0.3
AIDE two_stage (U3)	34	51.2	13.1	1.4	0	0.3
Mean patients treated						
BOIN12	10	13.6	4.7	1.3	0.2	
U-BOIN	10.8	12.1	5.1	1.5	0.4	
EffTox	26.6	1.8	0.9	0.4	0.1	
AIDE one_stage (U2)	9.2	14.4	5	1.1	0.1	
AIDE one_stage (U3)	9.7	14.8	4.4	1	0.1	
AIDE two_stage (U2)	10.2	9.9	7.2	2.3	0.4	
AIDE two_stage (U3)	10.2	10	7.1	2.3	0.4	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
BOIN12	15.8	19.6	21	15.5	28.1	0
U-BOIN	7.9	10.8	18.2	15.9	47.2	0
EffTox	6	7	21	18	42	6
AIDE one_stage (U2)	11.8	10.3	8.4	13.5	56	0
AIDE one_stage (U3)	18.3	14.2	9	11.9	46.6	0
AIDE two_stage (U2)	10.5	8.1	10.2	14.3	56.1	0.8
AIDE two_stage (U3)	16.3	12.2	9	13.8	48	0.7
Mean patients treated						
BOIN12	4.2	5.2	6	6.7	7.9	
U-BOIN	4.2	5.4	6.8	7.1	6.8	
EffTox	8.5	3.9	6.2	4.1	6.5	
AIDE one_stage (U2)	3.5	4.1	5.3	6.9	10.2	
AIDE one_stage (U3)	3.6	4.2	5.3	7	10	
AIDE two_stage (U2)	3.7	4	4.8	5.9	11.6	
AIDE two_stage (U3)	3.7	4	4.8	5.8	11.6	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
BOIN12	26.7	57	11.2	0.7	0	4.4
U-BOIN	16.5	39.9	22.5	11.5	9.3	0.3
EffTox	45	28	9	6	3	10
AIDE one_stage (U2)	16.9	59.7	17.6	2.3	0	3.5
AIDE one_stage (U3)	18.6	59	16.7	1.9	0	3.8
AIDE two_stage (U2)	19.5	54.5	15.5	1.7	0.1	8.7
AIDE two_stage (U3)	23.6	53.1	13.2	1.2	0.1	8.8
Mean patients treated						
BOIN12	10	12.1	5.4	1.8	0.4	
U-BOIN	7.8	10.3	6.3	3.4	2.1	
EffTox	17.6	5.9	3.2	1.3	0.6	
AIDE one_stage (U2)	8.2	12.7	6.3	2.2	0.4	
AIDE one_stage (U3)	8.5	12.8	6.1	2	0.3	
AIDE two_stage (U2)	9.2	10.5	6.9	2	0.3	
AIDE two_stage (U3)	9.2	10.4	7	2.1	0.3	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
BOIN12	16.7	58	18.5	5.2	0.7	0.9
U-BOIN	10	45	16.8	14	14	0.2
EffTox	24	30	20	7	10	9
AIDE one_stage (U2)	9.1	50.1	27.6	10.9	1.8	0.5
AIDE one_stage (U3)	13	51.4	24	9.8	1.3	0.5
AIDE two_stage (U2)	13.1	45.9	26.8	8.6	1.2	4.4
AIDE two_stage (U3)	19.4	48.3	20.7	6.6	0.7	4.3
Mean patients treated						
BOIN12	6.7	10.7	6.8	4.1	1.7	
U-BOIN	6.1	10.8	6.8	4	2.3	
EffTox	13.5	6.6	5.1	2	1.7	
AIDE one_stage (U2)	4.8	10.4	8.2	4.9	1.7	
AIDE one_stage (U3)	5.2	10.6	7.8	4.6	1.6	
AIDE two_stage (U2)	5.6	9	8.4	4.5	1.9	
AIDE two_stage (U3)	5.6	9	8.4	4.5	2	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
BOIN12	58.5	0.9	0.1	0	0	40.5
U-BOIN	49.7	8.4	7.2	4.9	5.8	24
EffTox	72	0	0	0	0	28
AIDE one_stage (U2)	57.1	1.8	0	0	0	41.1
AIDE one_stage (U3)	57.2	1.6	0	0	0	41.2
AIDE two_stage (U2)	60.2	1.4	0	0	0	38.4
AIDE two_stage (U3)	60.4	1.1	0	0	0	38.5
Mean patients treated						
BOIN12	19.9	2.6	0.4	0	0	
U-BOIN	17	4	1.6	1	0.8	
EffTox	25	0.3	0.1	0	0	
AIDE one_stage (U2)	20.2	2.5	0.2	0	0	
AIDE one_stage (U3)	20.3	2.4	0.2	0	0	
AIDE two_stage (U2)	20.9	2.1	0.6	0	0	
AIDE two_stage (U3)	20.9	2.1	0.6	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
BOIN12	40.8	50.2	3.8	0.6	0	4.6
U-BOIN	25.2	30.3	10	23.2	11.1	0.2
EffTox	39	29	10	4	2	16
AIDE one_stage (U2)	17.8	65.1	12.4	2.6	0	2.1
AIDE one_stage (U3)	22.2	62.3	10.8	2.7	0	2
AIDE two_stage (U2)	26.2	55.9	12.1	1.2	0	4.6
AIDE two_stage (U3)	29.1	55.4	9.3	1.6	0	4.6
Mean patients treated						
BOIN12	11.6	11.4	4.4	2	0.3	
U-BOIN	8.3	9.7	5	4.6	2.4	
EffTox	17	5.9	3.3	1.2	0.5	
AIDE one_stage (U2)	8.8	13.3	5.5	2.1	0.2	
AIDE one_stage (U3)	9.5	13	5.3	2	0.2	
AIDE two_stage (U2)	8.9	12.2	6.3	1.6	0.2	
AIDE two_stage (U3)	8.9	12.2	6.3	1.7	0.2	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
BOIN12	24.5	3	0	0	0	72.5
U-BOIN	47.6	18.1	5.9	1.9	0.5	26
EffTox	5	4	4	1	0	86
AIDE one_stage (U2)	20.4	4	0	0	0	75.6
AIDE one_stage (U3)	19.9	4.1	0	0	0	76
AIDE two_stage (U2)	18.5	5	0.2	0	0	76.3
AIDE two_stage (U3)	19.7	4.2	0	0	0	76.1
Mean patients treated						
BOIN12	16.1	4.7	1.1	0.1	0	
U-BOIN	13.3	4.7	2.9	2.6	2.6	
EffTox	8	3.1	1.7	0.3	0	
AIDE one_stage (U2)	15.8	4.6	0.7	0.1	0	
AIDE one_stage (U3)	15.8	4.5	0.7	0.1	0	
AIDE two_stage (U2)	17.2	2.6	0.6	0	0	
AIDE two_stage (U3)	17.2	2.7	0.6	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
BOIN12	2.6	2.1	6.9	62.8	25.6	0
U-BOIN	1.2	1.7	7.3	60.4	29.4	0
EffTox	5	5	10	18	63	1
AIDE one_stage (U2)	2.3	1.8	3	56.2	36.7	0
AIDE one_stage (U3)	4.4	3.2	4.1	58.3	30	0
AIDE two_stage (U2)	1.2	1.4	5.6	45	46.8	0
AIDE two_stage (U3)	2.5	2.5	7.2	48.7	39.1	0
Mean patients treated						
BOIN12	3.2	3.4	4.5	12.5	6.3	
U-BOIN	3.4	3.9	5	9.7	8.3	
EffTox	8.1	3.3	4.6	4	9.9	
AIDE one_stage (U2)	3.1	3.2	4	12.8	7	
AIDE one_stage (U3)	3.1	3.2	4.1	13.2	6.4	
AIDE two_stage (U2)	3.3	3.3	3.8	8.6	11.1	
AIDE two_stage (U3)	3.3	3.3	3.8	8.6	11.1	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
BOIN12	43.2	36	20.1	0.6	0	0.1
U-BOIN	30	25.3	26.2	9.9	8.5	0.1
EffTox	77	12	5	3	1	3
AIDE one_stage (U2)	29.2	32.5	33.1	5.1	0	0.1
AIDE one_stage (U3)	34	32.4	28.4	5	0	0.2
AIDE two_stage (U2)	34	29.3	32.4	3.6	0	0.7
AIDE two_stage (U3)	40	29.2	26.9	3.3	0	0.6
Mean patients treated						
BOIN12	10.2	9.8	7.5	2.2	0.3	
U-BOIN	9.7	9	7.2	2.9	1.2	
EffTox	22.9	3.3	2	1	0.3	
AIDE one_stage (U2)	9	9.3	8.7	2.7	0.4	
AIDE one_stage (U3)	9.7	9.3	8.3	2.4	0.3	
AIDE two_stage (U2)	9.3	8.3	9.3	2.6	0.4	
AIDE two_stage (U3)	9.4	8.2	9.3	2.6	0.4	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
BOIN12	31.1	32.3	32.3	0.9	0	3.4
U-BOIN	15.8	22.3	36.7	12.9	11.2	1.1
EffTox	38	25	18	8	4	8
AIDE one_stage (U2)	25	32.9	31.7	6.1	0.1	4.2
AIDE one_stage (U3)	27.9	35.6	27.1	5.2	0.1	4.1
AIDE two_stage (U2)	29.5	28.2	26.2	5.6	0.1	10.4
AIDE two_stage (U3)	33.6	27.5	23.4	5	0	10.5
Mean patients treated						
BOIN12	9.4	8.9	8.1	2.9	0.4	
U-BOIN	7.9	7.6	8	4.1	2.1	
EffTox	17	4.9	3.8	2.1	1	
AIDE one_stage (U2)	9.1	9.1	8.1	3	0.4	
AIDE one_stage (U3)	9.6	9	7.9	2.8	0.4	
AIDE two_stage (U2)	11	6.7	8	2.8	0.4	
AIDE two_stage (U3)	11	6.7	8	2.8	0.4	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
BOIN12	22.1	32.1	40.3	3.5	0.2	1.8
U-BOIN	9.9	20.2	39.6	15.7	14.4	0.2
EffTox	36	27	15	12	6	4
AIDE one_stage (U2)	16.3	27.6	44.6	10.4	0.4	0.7
AIDE one_stage (U3)	19	28.5	41.5	9.8	0.3	0.9
AIDE two_stage (U2)	21	23.9	41	7.6	1	5.5
AIDE two_stage (U3)	25.9	23.1	38.5	6.7	0.4	5.4
Mean patients treated						
BOIN12	7.5	9	9.1	3.3	0.8	
U-BOIN	6.9	7.9	8.9	4.4	1.8	
EffTox	16.6	5.1	3.7	2.2	1.7	
AIDE one_stage (U2)	6.7	8.5	10.1	3.9	0.8	
AIDE one_stage (U3)	7.2	8.3	10	3.7	0.7	
AIDE two_stage (U2)	8.1	6.7	9.5	3.9	1.2	
AIDE two_stage (U3)	8.1	6.6	9.5	4	1.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
BOIN12	25.1	50.6	4.4	0.5	0	19.4
U-BOIN	16.7	39.7	15.4	12.7	11.3	4.2
EffTox	36	29	8	5	4	19
AIDE one_stage (U2)	20.2	50.3	7.7	0.2	0	21.6
AIDE one_stage (U3)	20.7	49.3	7.4	0.3	0	22.3
AIDE two_stage (U2)	22.4	37	5.5	0.3	0	34.8
AIDE two_stage (U3)	22.9	37.1	5	0.2	0	34.8
Mean patients treated						
BOIN12	11.5	11.1	4	1.3	0.2	
U-BOIN	8.4	9	5.5	3.5	2.5	
EffTox	15.7	6.1	3.3	1.4	0.7	
AIDE one_stage (U2)	11	11.9	3.9	1.1	0.1	
AIDE one_stage (U3)	11.4	11.8	3.7	1.1	0.1	
AIDE two_stage (U2)	12.2	8.6	4.1	0.8	0.1	
AIDE two_stage (U3)	12.1	8.6	4.1	0.8	0.1	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
BOIN12	71	24.8	1.2	0	0	3
U-BOIN	61.5	24.3	4.9	3.8	3.6	1.9
EffTox	96	1	0	0	0	3
AIDE one_stage (U2)	64.5	30.1	2.8	0.2	0	2.4
AIDE one_stage (U3)	67.3	28	2.3	0.2	0	2.2
AIDE two_stage (U2)	67.9	24.9	4.2	0.2	0	2.8
AIDE two_stage (U3)	71.1	22.8	3.4	0.1	0	2.6
Mean patients treated						
BOIN12	17.1	8.8	2.8	0.6	0	
U-BOIN	16.2	8.8	3	1	0.6	
EffTox	28.3	0.7	0.3	0.1	0	
AIDE one_stage (U2)	17.7	8.8	2.5	0.5	0	
AIDE one_stage (U3)	18.5	8.5	2.1	0.4	0	
AIDE two_stage (U2)	18	7.4	3.5	0.5	0	
AIDE two_stage (U3)	18	7.4	3.6	0.5	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
BOIN12	22.4	21.9	19.1	33.7	2.9	0
U-BOIN	12.4	14.7	17.5	38.8	16.6	0
EffTox	39	14	20	14	11	3
AIDE one_stage (U2)	17.4	13.4	12.4	44.5	12.3	0
AIDE one_stage (U3)	25.6	15	11.1	38.5	9.8	0
AIDE two_stage (U2)	21.5	14.6	15.2	36.3	11.6	0.8
AIDE two_stage (U3)	28	15.9	13.1	33.2	9	0.8
Mean patients treated						
BOIN12	5.2	6.1	6.7	8.8	3.2	
U-BOIN	5.4	6.6	7.3	7.4	3.7	
EffTox	15.9	3.5	4.6	2.8	2.8	
AIDE one_stage (U2)	4.4	5	6.1	10.3	4.3	
AIDE one_stage (U3)	4.6	5.1	6.1	10.3	3.9	
AIDE two_stage (U2)	5.2	4.6	5.9	8.3	5.9	
AIDE two_stage (U3)	5.2	4.6	5.9	8.3	5.9	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
BOIN12	3.1	41.6	34.4	20.1	0.8	0
U-BOIN	4.6	39.7	34.4	20	1.3	0
EffTox	62	9	15	12	3	0
AIDE one_stage (U2)	4.4	35.8	29.1	25	5.7	0
AIDE one_stage (U3)	6.3	42	27.8	19.8	4.1	0
AIDE two_stage (U2)	8	35.6	25.6	24.5	6.2	0.1
AIDE two_stage (U3)	10.3	42.9	23.5	19.7	3.5	0.1
Mean patients treated						
BOIN12	3.8	12.3	8.1	4.6	1.1	
U-BOIN	4.5	7.6	7.8	7.5	3	
EffTox	20.7	2.9	3.1	2.3	1	
AIDE one_stage (U2)	3.8	11.7	7.4	5.5	1.6	
AIDE one_stage (U3)	3.8	12.3	7.3	5.2	1.4	
AIDE two_stage (U2)	4.1	6	6.3	7.8	5.8	
AIDE two_stage (U3)	4.1	6.1	6.3	7.8	5.8	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
BOIN12	14.5	20.5	46.3	15	3.7	0
U-BOIN	6.5	16.9	45.1	17.9	13.6	0
EffTox	19	18	30	14	15	4
AIDE one_stage (U2)	13.2	15.3	33.2	26.1	12.2	0
AIDE one_stage (U3)	18.8	16.8	33.8	22.7	7.9	0
AIDE two_stage (U2)	9.8	14.1	34.8	26.9	13.8	0.6
AIDE two_stage (U3)	15.7	18.1	33.1	24.6	7.9	0.6
Mean patients treated						
BOIN12	4.3	5.9	9.5	6.7	3.6	
U-BOIN	4.4	6	9.8	6.6	3.4	
EffTox	12.1	4.7	5.9	3.6	3.2	
AIDE one_stage (U2)	3.5	4.4	8.6	8.5	5	
AIDE one_stage (U3)	3.6	4.6	8.9	8.4	4.5	
AIDE two_stage (U2)	4	4.5	7.7	7.3	6.5	
AIDE two_stage (U3)	3.9	4.5	7.7	7.3	6.4	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
BOIN12	14.7	15.1	28.2	30.3	11.7	0
U-BOIN	6	7.9	27.8	28.5	29.8	0
EffTox	6	9	34	21	29	1
AIDE one_stage (U2)	8.4	5.6	11.6	44.8	29.6	0
AIDE one_stage (U3)	13	8.3	10.8	44.1	23.8	0
AIDE two_stage (U2)	9.6	4.7	14.7	40.8	30.2	0
AIDE two_stage (U3)	14.5	6.8	14.2	40.6	23.8	0.1
Mean patients treated						
BOIN12	4.5	5.6	7.3	8.2	4.4	
U-BOIN	3.9	5	8.7	8	4.5	
EffTox	8.9	4.1	7.1	4.5	5.3	
AIDE one_stage (U2)	3.4	3.7	6.2	10.5	6.2	
AIDE one_stage (U3)	3.5	3.8	6.2	10.6	5.9	
AIDE two_stage (U2)	3.5	3.7	5.4	8.6	8.8	
AIDE two_stage (U3)	3.5	3.6	5.4	8.5	8.9	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
BOIN12	28.2	22.3	35.3	10.8	2.8	0.6
U-BOIN	10.7	10.4	28.2	23	27.7	0
EffTox	12	23	27	11	16	10
AIDE one_stage (U2)	14.8	14.4	44.6	21.4	4.4	0.4
AIDE one_stage (U3)	18.7	16.6	43.2	18.3	2.7	0.5
AIDE two_stage (U2)	14	11.7	49	17.6	5.7	2
AIDE two_stage (U3)	20.8	15.1	43.1	15.9	3.2	1.9
Mean patients treated						
BOIN12	7.4	8	7.6	4.7	2.3	
U-BOIN	5.8	7.4	8.1	5.3	3.5	
EffTox	10.1	5.9	6.4	2.9	3.4	
AIDE one_stage (U2)	5.2	6.4	9.7	6.1	2.5	
AIDE one_stage (U3)	5.7	6.4	9.7	6	2.3	
AIDE two_stage (U2)	5.3	6	10	5.5	3	
AIDE two_stage (U3)	5.3	5.9	10	5.5	3	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
BOIN12	10.5	62.5	21.7	4.1	1.2	0
U-BOIN	9.9	61.2	20.2	5.9	2.8	0
EffTox	66	21	8	4	2	0
AIDE one_stage (U2)	8.5	42.4	31.4	14.5	3.2	0
AIDE one_stage (U3)	12.6	46.4	27.5	11.1	2.4	0
AIDE two_stage (U2)	11.7	42	31.3	11.4	3.5	0.1
AIDE two_stage (U3)	17	44.2	26.1	10.5	2.1	0.1
Mean patients treated						
BOIN12	5.3	13.8	6.8	2.9	1.1	
U-BOIN	5.5	12.8	7.8	3	1	
EffTox	21.4	4.3	2.7	1.1	0.4	
AIDE one_stage (U2)	4.4	11.7	8.4	4.2	1.4	
AIDE one_stage (U3)	4.6	12.4	7.9	3.9	1.2	
AIDE two_stage (U2)	4.5	8.7	8.8	5.2	2.7	
AIDE two_stage (U3)	4.6	8.7	8.8	5.1	2.8	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
BOIN12	29.9	38.9	13.5	6.3	2.8	8.6
U-BOIN	9.3	17.6	18.3	22.9	31.4	0.5
EffTox	22	20	15	7	18	19
AIDE one_stage (U2)	21.2	47.7	14.7	6.8	1.8	7.8
AIDE one_stage (U3)	22.8	50	10.8	7.4	1.1	7.9
AIDE two_stage (U2)	21.1	41.6	14	6	1.4	15.9
AIDE two_stage (U3)	25.8	40.8	10.3	5.9	1	16.2
Mean patients treated						
BOIN12	9.7	9.1	5.1	3.7	1.8	
U-BOIN	8	7.3	5.6	5	4.1	
EffTox	13.3	5.3	4	2	2.5	
AIDE one_stage (U2)	8.9	10.5	5.5	3.3	1.1	
AIDE one_stage (U3)	9.5	10.3	5.3	3.2	1	
AIDE two_stage (U2)	9.8	8.3	5.8	2.8	1.1	
AIDE two_stage (U3)	9.7	8.3	5.8	2.8	1.2	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
BOIN12	39.2	39	8.9	1.5	0.5	10.9
U-BOIN	25.3	25.9	13.7	19.8	15.2	0.1
EffTox	22	26	15	6	7	25
AIDE one_stage (U2)	20.6	57.6	14.1	3.1	0.7	3.9
AIDE one_stage (U3)	24.5	56.4	12	2.6	0.6	3.9
AIDE two_stage (U2)	27.2	49.8	12.4	2.9	0.2	7.5
AIDE two_stage (U3)	31.8	48.2	9.9	2.5	0.2	7.4
Mean patients treated						
BOIN12	11.2	10.1	4.5	2.4	1.1	
U-BOIN	8.7	8.4	5.3	4.3	3.4	
EffTox	13.6	6	4.2	1.5	1.2	
AIDE one_stage (U2)	8.7	11.8	5.5	2.9	0.8	
AIDE one_stage (U3)	9.4	11.5	5.4	2.8	0.8	
AIDE two_stage (U2)	8.9	10.6	6.2	2.3	0.8	
AIDE two_stage (U3)	8.9	10.5	6.3	2.3	0.7	